

ADAMAWA STATE UNIVERSITY, MUBI

FACULTY OF SCIENCE

DEPARTMENT OF COMPUTER SCIENCE

End of First Semester 2023/2024 Session Examination

CSC 401: Introduction to Artificial Intelligence

3 Units

Instruction: Attempt Question ONE and any other 3 Questions

Time: 2 hours

- | | |
|--|-------------|
| 1a. What is the most important characteristics of an expert system and why? | [05 Marks] |
| 1bi. What is knowledge? | [03 Marks] |
| ii. Briefly explain the components of a knowledge base | [03 Marks] |
| 1c. What is rationality? | [03 Marks] |
| 2a. Identify the unwelcomed properties of knowledge in the real world | [03 Marks] |
| 2b. Outline how human intelligence is different from machine intelligence | [06 Marks] |
| 2c. Outline two broad types of reasoning as components of Artificial Intelligence | [05 Marks] |
| 3a. Precisely state what it takes to be Intelligent | [04 Marks] |
| 3b. What are the conditions on which rational agent depends? | [04 Marks] |
| 3ci. When does a machine said to be a robot? | [06 Marks] |
| 4a. Outline four parts of a Biological Neural Network | [06 Marks] |
| 4b. Use diagrams only to illustrate the three categories of learning in ANN | [08 Marks] |
| 5a. Given that a robot has 4 legs, what is the number of possible events of the robot? | [03½ Marks] |
| 5b. Outline the types of intelligence as described by Howard Gardner. | [10½ Marks] |
| 6a. Outline the limitations of Genetic Algorithms | [08 Marks] |
| 6b. Why it is necessary to study AI. | [03 Marks] |
| 4c. Outline three usefulness of fuzzy logic | [03 Marks] |



ADAMAWA STATE UNIVERSITY, MUBI
FACULTY OF SCIENCE
DEPARTMENT OF COMPUTER SCIENCE

END OF FIRST SEMESTER EXAMINATION 2023/2024 SESSION

COURSE CODE: CSC 403 UNITS: 03

COURSE TITLE: INTRODUCTION TO SOFTWARE ENGINEERING

INSTRUCTION: ATTEMPT ANY FOUR QUESTIONS.

TIME ALLOWED: 3 HOURS

1. (a) Define the following terms as they are used in software Engineering:
 - i. Software. (2 Mk)
 - ii. Engineering. (2 Mk)
 - iii. Software Engineering. (2Mk)(b) In not more than five sentences, state why you think the knowledge of software engineering is required in computing discipline. (5 Mks)
(c) i Outline any four importance of software engineering. (2 Mks)
ii. Outline the difference between object-oriented design and function-oriented design. (4.5Mks)
2. (a) i. A software process is a set of activities and associated outcome that produce a software product. Mostly these activities are carried out by software engineers. Outline these activities. (4 Mks)
ii. In software Engineering, what do you think are the causes of software crises? (4 Mks)
(b) What differentiates software documentation from operating procedure? (5.5 Mks)
(c) i. As a software Engineer, state the circumstances where the use of waterfall model of software development is most suited for a project.(2 Mks)
ii. State any two of the drawbacks of the model in (c) i above. (2 Mks)
3. (a) Outline the difference between spiral model and iterative model of software development. (6 Mks)
(b) List any four Agile methodology testing methods and explain any two of them. (7 Mks)
(c) A good software requirement specification document has some essential properties, outline any three of these properties (4.5 Mks)
4. (a) As a software engineer, you are saddled with the responsibility to create reasons for developing a software that is acceptable to user, flexible to changes and conformable to established standard. What do you think are the types of feasibility you need to carry out? (6.5 Mks)
(b) State six problems of elicitation and analysis in software development process. (6 Mks)
(c) State any five (5) pros of agile software development methodology. (5 Mks)

5. (a) i. What do you understand by the term software design? State its basic objectives. (4.5 Mks)
ii. Highlight any five of the potential benefits of modularity in software Engineering. (2.5 Mks)
- (b) As a good code writer, outline the basic principles of software design. (4 Mks)
- (c) i. In modular design, how do you achieve functional independence and what is the basic criteria of measuring it? (3 Mks)
ii. In programming, what you understand by module cohesion. (3.5 Mks)
6. (a) What do you understand by the concept of the Application Programming Interface (API)? (5 Mks)
- (b) Using diagram, explain the working concept of (a) above. (8.5 Mks)
- (c) Enumerate the challenges developers faced when using API. (4Mks)

ADAMAWA STATE UNIVERSITY, MUBI

FACULTY OF SCIENCE

DEPARTMENT OF COMPUTER SCIENCE

END OF FIRST SEMESTER 2021/2022 SESSION EXAMINATIONS

COURSE CODE/TITLE: CSC 411/ OOP PROGRAMMING USING JAVA

Instructions: ANSWER Question 1 and any other three (3) QUESTIONS. TIME: 3 HOURS

Q1. (a) Articulate with an example the process of referencing object's members in Java. (12marks)

(b) Discuss the essential feature of Java that allows a single compiled program to be executed on different machine architecture configurations. (8marks)

(c) Explain why classes are regarded as datatypes in OOP. (5marks)

Q2. (a) Explain the syntax and semantics of every statement in the program below. (10 Marks)

```
import java.util.Scanner;

public class ScannerDemol {

    public static void main(String[] args)
    {

        Scanner sc = new Scanner(System.in);
        String name = sc.nextLine();
        char gender = sc.next().charAt(0);
        int age = sc.nextInt();
        long mobileNo = sc.nextLong();
        double cgpa = sc.nextDouble();

        System.out.println("Name: " + name);
        System.out.println("Gender: " + gender);
        System.out.println("Age: " + age);
        System.out.println("Mobile Number: " + mobileNo);
        System.out.println("CGPA: " + cgpa);

    }

}
```

(b) Write down the output of the program in question 2(a) above. (5 Marks)

Q3. (a) Why do we need control structures in programming? (3 marks)

(b) Discuss with examples the syntax for any three (3) control structures in java. (9 Marks)

(c) Discuss the differences between while and do-while in Java? (3 marks)

Q4. (a) Explain the switch statement in java and its rules thereof. (5 marks)

(b) Imagine having a selection guaranteed from a variable whose possible outcomes are five (5) integers, write a switch block to implement the selection. (10 marks)

Q5. (a) With an implementable example, discuss why we consider arrays as an object in Java programming? (7 Marks)

(b) Implement a program that can assign and retrieve integer values for a one dimensional array of size 17. (8 Marks)

Q6. (a) Why it is necessary to have constructors in Java classes? (5 Marks)

(b) Write a java class with two (2) constructors, one parameterized and the other non-parameterized. (10 Marks)

ADAMAWA STATE UNIVERSITY
(FACULTY OF SCIENCE)

DEPARTMENT OF COMPUTER SCIENCE

First Semester Examinations for 2023/2024 Session

Course: CSC 409 Operating Systems II (3 Units)

Time Allowed: 3 Hours

Full Marks: 70%

INSTRUCTIONS: ANSWER ANY FOUR (4) QUESTIONS

No. 1

- State the steps upon which the disk performance requirement depends. **4.5 Marks**
- Explain the two writing policies a file system employs when caching significant amount of data in memory? State the advantage and disadvantage of each. **7.0 Marks**
- Consider a disk system with a sector size of 512 bytes. If a file system block is 4 KB, how many contiguous sectors does the block contain? What are the sectors_ids for block numbers 3, 5 and 7? **6.0 Marks**

No. 2

- A particular computer system has 8 GB of RAM and the processor can issue a 34-bit virtual address. If the offset is 13 bits wide and a user program of 129,024 bytes, determine the following.
 - The size of each page and frame **1.5 Marks**
 - The number of pages the virtual memory can support **1.5 Marks**
 - The number of frames the physical memory can support **1.5 Marks**
 - The number of frames this program would require, assuming the program can be fully loaded into RAM. **1.5 Marks**
- Consider a virtual memory system with the following parameters: 6 pages, size of each page is 4 bytes and the following page mapping to frames.

Pages	Frames
0	7
1	5
2	4
3	3
4	6
5	0

Determine the physical addresses to which the following logical addresses map: (i) 6, (ii) 10, (ii) 16 and (iv) 20 **6.0 Marks**

- What is concurrency and how does it differ from parallelism? What are the issues and difficulties raised by concurrency? **5.5 Marks**

No. 3

- a. What is process synchronisation? In one sentence state critical section problem. **5.0 Marks**
- b. What is deadlock? Using a resource allocation graph show how four processes can be deadlocked if there are only three resources. One Resource has two instances and the rest has one instance each. List the cycles that caused the processes to be deadlocked. **6.5 Marks**
See graph this is from b.
- c. What is scheduling? List the conditions under which scheduling decisions may take place and give examples for each. **6.0 Marks**

No. 4

- a. Briefly explain the following page replacement algorithms: (i) FIFO, (ii) Optimal and (iii) LRU **4.5 Marks**
- b. Given the following page reference string 0, 2, 1, 6, 4, 0, 1, 0, 3, 1, 2, 1, determine the number of page faults for FIFO, Optimal and Least recently used algorithms if there are only four frames in the memory. **7.5 Marks**
- c. What is a critical section and how do they relate to controlling access to shared resources? What is race condition? **5.5 Marks**

No. 5

- a. In one sentence explain first-fit, best-fit and worst-fit dynamic allocation strategies. Suppose that we have free frames with sizes: 6, 17, 25, 14, and 19. Place a program with size 13kB in the free segment using first-fit, best-fit and worst-fit. **6.0 Marks**
- b. Discuss external and internal fragmentation of memory, and how do the dynamic allocation strategies affect them? **4.0 Marks**
- c. Consider OS process and process management and answer the following questions.
- i. Briefly discuss the various components of a process. **3.0 Marks**
 - ii. Draw a neat diagram of process state transitions. **2.0 Marks**
 - iii. What is context switch? **1.5 Marks**
 - iv. What is the size of memory in bytes that is available to processes to use on a machine with 34-bit address. **1.0 Marks**

No. 6

- a. The block size of a file system is 2 KB and the disk block numbers can be stored in 4 bytes. What is the maximum file size supported by a file system with 13 direct blocks, single, double, and triple indirection? **5.0 Marks**
- b. What is processor affinity in multicore scheduling? Discuss the two forms of processor affinity. **5.0 Marks**
- c. Consider rate-monotonic scheduling algorithm and answer the following questions.
- i. Briefly explain rate-monotonic scheduling algorithm in real-time systems. **3.0 Marks**
 - ii. Assuming we have two threads with the following details. T1 period = 40, T2 period = 90, T1 CPU burst time = 15 and T2 CPU burst time = 30. The deadline for each process requires that it complete its CPU burst by the start of its next period. Determine the total CPU utilization and show the execution of T1 and T2 on a time chart using rate-monotonic scheduling algorithm. **4.5 Marks**

15/02/2024
ADAMAWA STATE UNIVERSITY, MUBI

(FACULTY OF SCIENCE)

DEPARTMENT OF COMPUTER SCIENCE

END OF FIRST SEMESTER EXAMINATION, 2023/2024 SESSION

CSC 415: NET-CENTRIC COMPUTING

UNITS: 3

INSTRUCTION: Answer Four (4) Questions only

70 Marks

1. (a). Net-centric computing is a distributed computing where applications and data are exchanged among peers across a network on a demand basis. Explain the distributed system and discuss the components of a typical distributed computing system [14 marks]
(b). Big data is defined as data that is high in volume, velocity, and diversity; requiring new technologies and techniques to acquire, store, and analyze; and is used to improve decision making, give insight and discovery, and support and optimize processes. Critically examine how high-performance computing (HPC) addresses the computational challenges of big data processing [3.5 Marks]
2. (a). Explain with a typical example how cloud computing support computations processes in a mobile device [7.5 Marks]
(b). Explain a communication process in a typical client-server network environment [10 marks]
3. (a). Mobile computing plays an important part in educational processes around the world; however, developing nations have been unable to capitalize on the distinguishing characteristics of this technology, which allows for learning to take place anywhere and at any time, resulting in a paradigm change in the learning process. Examine seven (7) distinguishing characteristics of this technology that lead to anywhere and anytime computing [14 Marks]
(b). Cloud computing is an evolution of current technologies, rather than a revolution in information technology. Critically examine the statement [3.5 Marks]
4. (a). An IT consortium wishes to apply the computational capabilities of cloud computing to its in-house computational activities. However, the consortium is conscious of administrative costs and the security of firm assets. Explain in detail five (5) benefits and drawbacks of a recommended cloud computing model for this collaboration [15 Marks]
(b). How can a computer locate another computer on a network? [2.5 Marks]
5. (a). The Internet of Things (IoT) refers to a network of physical "things," embedded with sensors, software, and other technologies to connect and exchange data with other devices and systems over the internet. Explain "things" with specific examples. [7.5 Marks]
(b). List and discuss five (5) ways in which mobile information systems support globalization [10 Marks]
6. (a). How do HPC, Mobile computing, parallel computing, cloud computing and IoT contribute to developing a distributed computing system? [15 Marks]
(b). Why net-centric Computing? [2.5 Marks]

Good luck



ADAMAWA STATE UNIVERSITY, MUBI
FACULTY OF SCIENCE
COMPUTER SCIENCE DEPARTMENT
END OF FIRST SEMESTER EXAMINATION
2023/2024 ACADEMIC SESSION

COURSE TITLE/CODE: Compiler Construction and Design (CSC 407)

UNIT: 3

INSTRUCTION: Answer four (4) questions only

TIME: 3 hours

- 1(a) What is a Context Free Grammar? (5 marks)
- (b) Show whether or not the following grammar is ambiguous
 $S \rightarrow OB/1A$
 $A \rightarrow O/OS/1AA$
 $B \rightarrow 1/1S/OBB$ for 00110101 (5 marks)
- (c) What are the key Steps involved in the conversion of a high level language into a machine code? (7 ½ marks)
2. Consider the following grammar: GI;
 $S \rightarrow Sa$
 $S \rightarrow bS$
 $S \rightarrow c$
- (a) What kind of grammar is this? (5 marks)
- (b) If GI is ambiguous, explain what it means. (5 marks)
- (c) How are ambiguous grammars handled while constructing a compiler? (7 ½ marks)
- 3(a) Let $R = \{(S, C), (a, b), p, f\}$ where P consist
 $S \rightarrow aCa$
 $C \rightarrow aCa|b$
Find $L(G)$ as a proof that $L(G) = \{a^n b a^n | n > 1\}$ belongs to $L(G)$ (7 ½ marks)
- (b) Discuss the actions taken by every phase of the compiler on the following string: $A=B*C+D/E$ (10 marks)
- 4(a) Construct a Finite Automata (FA) that accepts an even number of zeroes (0s) and an even number of ones (1s) (5 ½ marks)
- (b) In the context of automata theory explain the following:
i. Recognized language ii. Recognizable language (1 ½ marks each)
iii. Run iv. Accepting word
- (c) Give the left and right most derivation parse tree for aabbbaa using:
 $S \rightarrow aAS/a$
 $A \rightarrow SbA/SS/ba$ (6 marks)
- 5(a) What are the advantages of:
i. Intermediate code of generation during compilation (5 marks)
ii. NFA over DFA (5 marks)
- (b) Construct an NFA and DFA for $(a/b)^*abb$ (7 ½ marks)
- (c) Describe the role of a Linker and a Loader in compiler design.
- 6(a) Mainly, compilers are divided into two (2) largely independent parses: Front-end and Back-end. Explain the difference between them? (4 ½ marks)
- (b) Difference between Parser and Compiler. (3 marks)
- (c) i. What is a useless grammar (2 marks)
ii. State the conditions that makes a grammar symbol X to be useful? (3 marks)
iii. Find the reduced form of the following grammar, stating all the necessary assumptions
 $A \rightarrow xyz|Xyzz$
 $X \rightarrow Xz|xYx$
 $Y \rightarrow yYy|XZ$
 $Z \rightarrow Zy|z$ (5 marks)